

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? _____ S0/0/0 _____

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? _____ 129 _____

Does R2 show up as an OSPF neighbor on R1? _____ Yes _____

Does R2 show up as an OSPF neighbor on R3? _____ No _____

What does this information tell you?

_____ This indicates that R3 cannot form a direct OSPF adjacency with R2 over the S0/0/1 link because that interface on R2 remains passive. As a result, R3 must route traffic to the 192.168.2.0/24 network indirectly through R1. _____

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

_____ R2(config)# router ospf 1 _____

_____ R2(config-router)# no passive-interface s0/0/1 _____

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? _____ S0/0/1 _____

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

_____ 65 (64+1) _____

64: T1 serial link (cost = 100 Mbps / 1.544 Mbps = 64)

1: R2 Ethernet interface link (G0/0) cost 1 _____

Is R2 listed as an OSPF neighbor to R3? _____ Yes _____

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

___ Increase the OSPF reference bandwidth from its default 100 Mbps. The default causes OSPF to assign equal costs to interfaces faster than 100 Mbps, reducing path selection accuracy. A higher reference bandwidth enables finer cost differentiation and more optimal routing based on actual link speeds. _____

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

___ 192.168.3.0/24: R1 S0/0/1(100Mbps / 128kbps = 781) + R3 G0/0 = 781 + 1 = 782 _____

___ 192.168.23.0/30: R1 S0/0/1(100Mbps / 128kbps = 781) + R3 S0/0/1 (100Mbps / 1.544Mbps = 64) = 781 + 64 = 845 _____

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

___ Two serial links, each serial link cost of 100 Mbps / 128 kbps = 781. _____

192.168.23.0/24: [R1 S0/0/0 (100 Mbps / 128 kbps = 781) + R2 S0/0/1 (100 Mbps / 128 kbps = 781)] or [R1 S0/0/1 (100 Mbps / 128 kbps = 781) + R3 S0/0/1 (100 Mbps / 128 kbps = 781) = 781 + 781 = 1562] = 781 + 781 = 1562 _____

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

___ The route to the 192.168.3.0/24 network (R3's LAN) on R1 now goes through R2 because OSPF chooses the path with the lowest total metric. Cost of R1 Serial0/0/1 interface (the direct link to R3) is 1565/ the direct path's accumulated cost increases to 1566. The path via R2 has a lower total cost of 1563 (781 for R1 Serial0/0/0 + 781 for R2 Serial0/0/1 + 1 for R3 G0/0). _____

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

___ Controlling the router ID in OSPF is essential because it uniquely identifies routers for communication and database sync. Automatic selection (based on highest IP) can be unpredictable with interface changes, complicating troubleshooting and management. It also affects DR/BDR elections in multi-access networks. Changing IDs post-configuration disrupts the process, so proactive control maintains stability and scalability.

2. Why is the DR/BDR election process not a concern in this lab?

___ In our lab, the serial links used which are point-to-point links. Therefore, no DR/BDR election happened. DR/BDR election process happens on a multi-access network.

3. Why would you want to set an OSPF interface to passive?

___ Setting an OSPF interface to passive allows the router to advertise the connected network into the OSPF domain while preventing the exchange of Hello packets and formation of adjacencies on that interface. It also reduces unnecessary protocol traffic, conserves bandwidth and CPU resources, and improves security by advertising the network to its neighbors.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

___ The use of multiple areas in OSPF enables its hierarchical structure, allowing route summarization at area borders, reduce routing cost, and enhance scalability for large networks.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

___ The single OSPF router configuration command is 'network 10.0.0.0 0.255.255.255 area 0'.

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

___ The EIGRP route will be selected. EIGRP has the lowest default AD of 90, followed by OSPF at 110 and RIP at 120. Therefore, the EIGRP route with lowest AD will be selected.